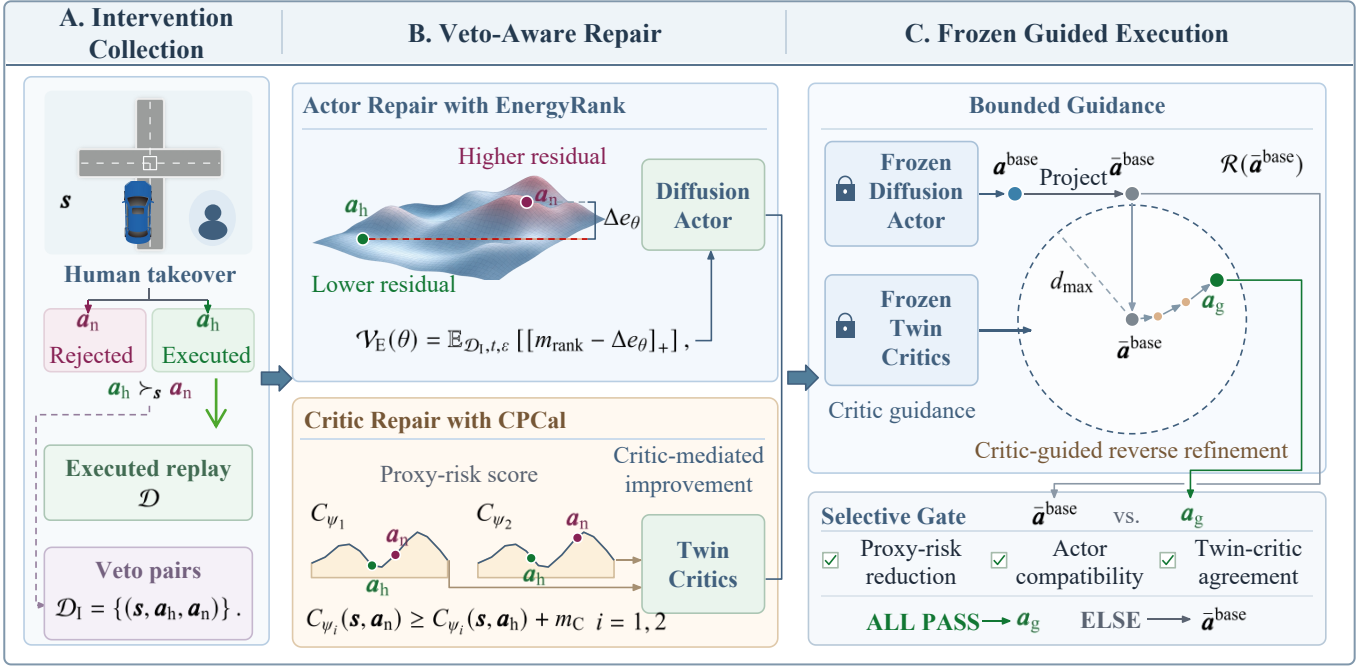

A. Intervention Collection
B. Veto-Aware Repair
C. Frozen Guided Execution
s
Human takeover
a_n
Rejected
a_h
Executed
$a_h \succ_s a_n$
Executed replay
$\mathcal{D}$
Veto pairs
$\mathcal{D}_I = \{(s, a_h, a_n)\}$.
Actor Repair with EnergyRank
Higher residual
a_h
a_n
Δe_θ
Lower residual
Diffusion Actor
$\mathcal{V}_E(\theta) = \mathbb{E}_{\mathcal{D}_I,t,\varepsilon}\left[[m_{\text{rank}} - \Delta e_\theta]_+\right]$,
Critic Repair with CPCal
Proxy-risk score
Critic-mediated improvement
C_{ψ_1}
C_{ψ_2}
Twin Critics
$C_{\psi_i}(s, a_n) \geq C_{\psi_i}(s, a_h) + m_C \quad i = 1, 2$
Bounded Guidance
Frozen Diffusion Actor
a^{base}
Project
$\bar{a}^{\text{base}}$
$\mathcal{R}(\bar{a}^{\text{base}})$
Frozen Twin Critics
$d_{\max}$
a_g
Critic guidance
Critic-guided reverse refinement
Selective Gate
$\bar{a}^{\text{base}}$ vs. a_g
Proxy-risk reduction
Actor compatibility
Twin-critic agreement
ALL PASS → a_g
ELSE → $\bar{a}^{\text{base}}$